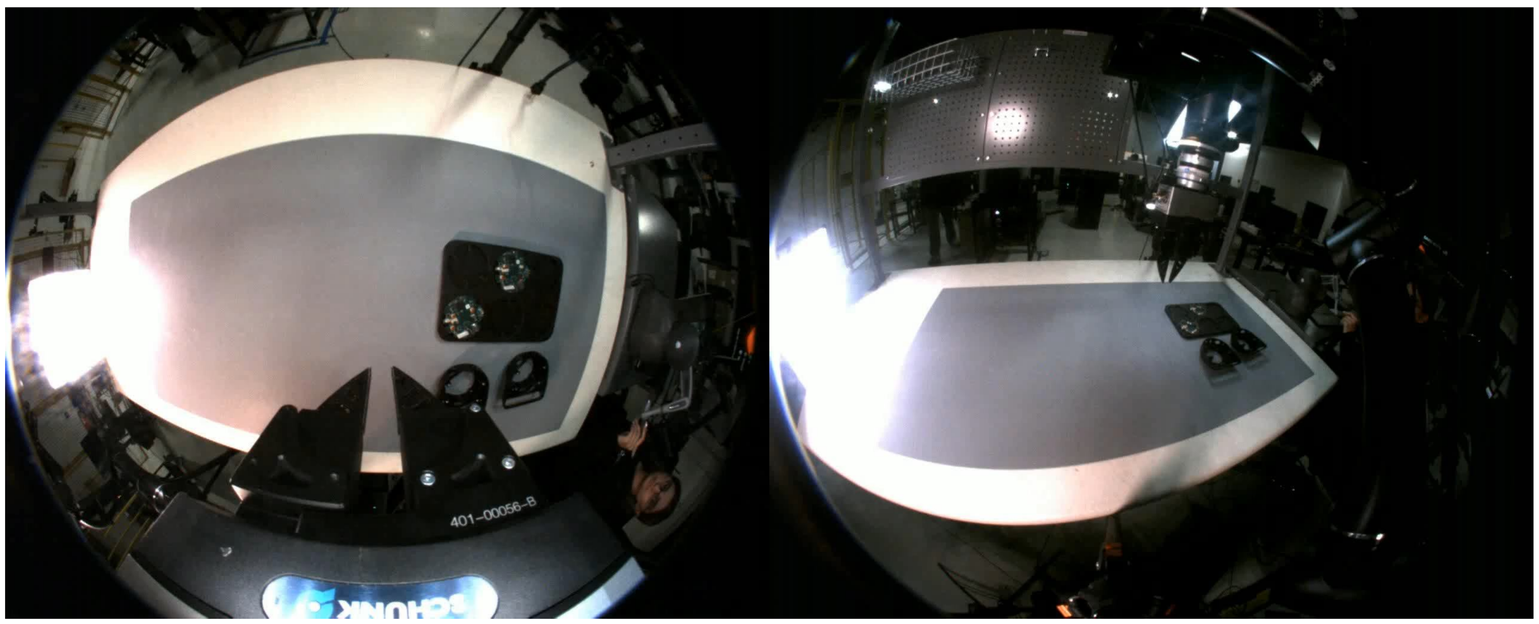
ML/AI Engineer — System Design Interview

Senior ML/AI Engineer (Robotics / Manipulation) · Open-ended system design · ~60 min

You are joining the team that builds learned manipulation **skills** for our programmable-robot platform.
Customers compose these skills into longer jobs through our planner; your job is to deliver one new skill, end-to-end.

The setup



Hardware · 6/7-DOF arm + parallel gripper · joint encoders · gripper state · optional F/T
Control · end-effector pose or joint targets at ~10–30 Hz
Sensing · cameras and other sensors are part of your design

Your task

Assemble the part and move completed part to designated area. We need to be able to handle unexpected movements / disturbances to the part and arbitrary placements of part. One part assembled per run.

AT MINIMUM, PLEASE COVER

1
Data
How you collect it, how much, and how you assess **quality** and **coverage**.

2
Model
What you'd train, why, and what you'd reject. We want to hear real tradeoffs.

3
Evaluation & iteration
How you know the policy is good enough, and how you decide what to do next.

4
Completion detection
Skill must tell the planner when it's **DONE** or has **FAILED**. Architecture, labels, and failure modes.

5
Operating in production
How the system gets better over time after deployment.

High level Components:

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1. Data collection (using tele op and other methods)
2. Injestion & Validation
3. Preprocessing & Format
4. Model Architectures
5. Training Infra
6. Evaluation
7. Deployment
8. Monitoring & Fylwheel